

5 Whys Analysis — AI-Powered Supply Chain & Procurement Analytics

Overview

Identify the underlying business/process causes behind major supply-chain problems and define improvement opportunities for the To-Be process.

Team	Supply Chain Analytics Project Team
Team members	Business Analyst, Supply Chain Manager, Procurement Manager, Data Analyst, Data Scientist/ML Analyst
Date	

Problem statements

Problem 1	Problem 2	Problem 3	Problem 4	Problem 5
Delivery Delays	Supplier Performance	Data Management	Inventory Risk	Inventory Risk
Whys	Whys	Whys	Whys	Whys
Delayed orders are identified too late, making it difficult to take early action.	Supplier performance is difficult to compare across delivery, delays, cost, and cancellations.	Supply-chain data requires manual cleaning and consolidation before analysis.	Potential inventory risks are difficult to identify early.	Management mainly relies on historical reports instead of proactive predictive insights.
Final problem statement				
The supply-chain process lacks integrated visibility, standardized analysis, and predictive insights, making it difficult to identify risks early and support proactive, data-driven decisions.				

Solutions

Solution	Owner	Status	Action items
Create centralized supply-chain dashboard	Business Analyst / BI Developer	IN PROGRESS	Define KPIs and build Power BI dashboard
Implement supplier performance analysis	Business Analyst / Procurement Manager	IN PROGRESS	Define supplier KPIs and compare supplier performance
Standardize data preparation	Data Analyst	IN PROGRESS	Clean, validate, and transform supply-chain data
Implement delivery-risk prediction	Data Scientist / ML Analyst	IN PROGRESS	Develop and evaluate delivery-risk model
Establish inventory-risk analysis	Business Analyst / Data Analyst	IN PROGRESS	Define inventory-risk indicators and identify high-risk areas
Introduce demand forecasting	Data Scientist / ML Analyst	IN PROGRESS	Build forecast and evaluate forecast accuracy
Create management reporting	Business Analyst	IN PROGRESS	Develop executive summary, WBR, and MBR reporting
Implement continuous KPI monitoring	Supply Chain Manager	IN PROGRESS	Review KPIs, risks, and recommended actions regularly